Cluster build (non kubernetes) of AI LLM in single instance and distributed (multi node) configuration based on suse 16

3 node setup:
2 nodes: 256gb ram, 14 cpu, 500gb storage
1 node: 128g ram, 8 cpu, 2tb storage

Test configurations:
Single instance of the ai llm via cli and server mode with local browser gui and openweb gui
Distributed configuration utilizing 2 nodes for the brain with the 3rd node for processing and logging
Incremental/iterative build and expansion of features and configurations – ongoing progress

Post build assessment:
The distributed – 2 node configuration bottleneck was definitely the 1gb backend for the llm performance, having 10gb or even 4gb lacp configuration would definitely improve performance. That's a near future upgrade configuration to test.

The single instance performance was noticeably improved, as it wasn't going through the network backend for the distributed 2 node compute and relay traffic.

// pre-requisites of packages to install, some development tools

```
sles16-node1:/home/onemantech # zypper install -t pattern devel_basis
Refreshing service 'openSUSE'.
Loading repository data...
Reading installed packages...
Resolving package dependencies...

The following 20 recommended packages were automatically selected:
  binutils-devel bison-lang cyrus-sasl-devel e2fsprogs-devel gcc-c++ gcc-info
  git git-email glibc-info gmp-devel gperf libaio-devel libdb-4_8-devel
  libosip2-devel libstdc++-devel makeinfo-lang openldap2_6-devel pam-devel patch
  sparse
```

```
sles16-node1:/home/onemantech # zypper install git-core cmake-full
Refreshing service 'openSUSE'.
Loading repository data...
Reading installed packages...
'git-core' is already installed.
No update candidate for 'git-core-2.51.0-160000.1.2.x86_64'. The highest available version is already installed.
Resolving package dependencies...

The following 4 NEW packages are going to be installed:
  cmake-full libjsoncpp26 libjsoncpp26-x86-64-v3 librhash1

4 new packages to install.
```

// this was installed for the git clone functionality, to pull the llama.cpp files

```
sles16-node3:/opt/ai-lab # zypper install git-core
Refreshing service 'openSUSE'.
Loading repository data...
Reading installed packages...
Resolving package dependencies...

The following 2 NEW packages are going to be installed:
  git-core libsha1detectcoll1
```

//github clone of the repo for the specifics
//making a test directory to keep things sorted/organized

// the llama.cpp package allows for the distributed configuration of running the llm on multiple nodes via rpc, rather than a single node instance

```
sles16-node1:/home/onemantech # mkdir -p /opt/ai-lab
sles16-node1:/home/onemantech # cd /opt/ai-lab/
sles16-node1:/opt/ai-lab # git clone https://github.com/ggml-org/llama.cpp.git
Cloning into 'llama.cpp'...
remote: Enumerating objects: 81845, done.
remote: Counting objects: 100% (138/138), done.
remote: Compressing objects: 100% (126/126), done.
remote: Total 81845 (delta 55), reused 12 (delta 12), pack-reused 81707 (from 4)
Receiving objects: 100% (81845/81845), 310.43 MiB | 9.50 MiB/s, done.
Resolving deltas: 100% (58866/58866), done.
sles16-node1:/opt/ai-lab #
sles16-node1:/opt/ai-lab #
sles16-node1:/opt/ai-lab # cd llama.cpp/
sles16-node1:/opt/ai-lab/llama.cpp # cmake -B build -DGGML_RPC=ON
-- The C compiler identification is GNU 15.2.0
-- The CXX compiler identification is GNU 15.2.0
-- Detecting C compiler ABI info
-- Detecting C compiler ABI info - done
-- Check for working C compiler: /bin/cc - skipped
```

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// pulling the ai llm model
// running the rpc server so the host is seen as a compute node

```
sles16-node1:/opt/ai-lab/llama.cpp # ollama pull gemma3:27b-it-qat
pulling manifest
pulling ccc0cddac561: 100%                        18 GB
pulling 5be6630b666f: 100%                       359 B
pulling 3116c5225075: 100%                        77 B
pulling 870e9c20a913: 100%                       414 B
verifying sha256 digest
writing manifest
success
sles16-node1:/opt/ai-lab/llama.cpp # build/bin/rpc-server -p 50052 --host 0.0.0.0

!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
WARNING: Host ('0.0.0.0') is != '127.0.0.1'
         Never expose the RPC server to an open network!
         This is an experimental feature and is not secure!
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!

Starting RPC server v3.6.0
  endpoint       : 0.0.0.0:50052
  local cache    : n/a
Devices:
  CPU: Intel(R) Xeon(R) Silver 4116 CPU @ 2.10GHz (257738 MiB, 257738 MiB free)
```

// this should be done on all the nodes, edit the ollama.service file in systemctl
// added the service/environment section

```
sles16-node1:/home/onemantech # systemctl edit ollama.service
```



```
onemantech@sles16-node1:/home/onemantech

      /etc/systemd/system/ollama.service.d/.#override.conf34d931469de51be3
### Editing /etc/systemd/system/ollama.service.d/override.conf
### Anything between here and the comment below will become the contents of the>

[Service]
Environment="OLLAMA_HOST=0.0.0.0"

### Edits below this comment will be discarded


### /etc/systemd/system/ollama.service
# [Unit]
# Description=Ollama Service
# After=network-online.target
#
# [Service]
# ExecStart=/usr/local/bin/ollama serve
# User=ollama
# Group=ollama
# Restart=always
# RestartSec=3
                                [ Read 24 lines ]
^G Help        ^O Write Out   ^F Where Is    ^K Cut         ^T Execute     ^C Location
^X Exit        ^R Read File   ^\ Replace     ^U Paste       ^J Justify     ^/ Go To Line
```

// listing the directory of where the model (blobs) resides

```
sles16-node3:/opt/ai-lab/llama.cpp # ls -lsh /usr/share/ollama/.ollama/models/blobs
total 17G
4.0K -rw-r--r--. 1 ollama ollama  77 Mar  6 00:32 sha256-3116c52250752e00dd06b16382e952bd33c34fd79fc4fe3a5d2c77cf7de1b14b
4.0K -rw-r--r--. 1 ollama ollama 359 Mar  6 00:32 sha256-5be6630b666f007a127a456fb3958da5fbd7cf547204188bc4690fedc837af0b
4.0K -rw-r--r--. 1 ollama ollama 414 Mar  6 00:32 sha256-870e9c20a913cbec51bee4a59f0747ee39e0a0ee2cd8cfc26f6ad6ba9681a7fe
 17G -rw-r--r--. 1 ollama ollama 17G Mar  6 00:31 sha256-ccc0cddac56136ef0969cf2e3e9ac051124c937be42503b47ec570dead85ff87
```

// running the distributed (dual brain compute) cli instance of the model on the 3rd node used at the management while the first 2 nodes are the compute/brains

```
sles16-node3:/opt/ai-lab/llama.cpp # build/bin/llama-cli -m build/bin/gemma3-27b-it-qat.gguf --rpc 172.[redacted]:50052,172.[redacted]
189:50052 --n-gpu-layers 0

Loading model... /llama_model_load: error loading model: done_getting_tensors: wrong number of tensors; expected 1247, got 808
llama_model_load_from_file_impl: failed to load model
-llama_params_fit: encountered an error while trying to fit params to free device memory: failed to load model
-llama_model_load: error loading model: done_getting_tensors: wrong number of tensors; expected 1247, got 808
llama_model_load_from_file_impl: failed to load model
\common_init_from_params: failed to load model 'build/bin/gemma3-27b-it-qat.gguf'
srv    load_model: failed to load model, 'build/bin/gemma3-27b-it-qat.gguf'

Failed to load the model
```

// this was part of python to allow communication/interactions with the gemma model and scripts/open webui

```
sles16-node3:/opt/ai-lab/llama.cpp # pip install -r requirements.txt
DEPRECATION: Loading egg at /usr/lib/python3.13/site-packages/suse_lifecycle-0.9-py3.13.egg is deprecated. pip 25.1 will enforce
this behaviour change. A possible replacement is to use pip for package installation. Discussion can be found at https://github.c
om/pypa/pip/issues/12330
Looking in indexes: https://pypi.org/simple, https://download.pytorch.org/whl/cpu, https://download.pytorch.org/whl/nightly, http
s://download.pytorch.org/whl/cpu, https://download.pytorch.org/whl/nightly, https://download.pytorch.org/whl/cpu, https://downloa
d.pytorch.org/whl/nightly
Ignoring torch: markers 'platform_machine == "s390x"' don't match your environment
Ignoring torch: markers 'platform_machine == "s390x"' don't match your environment
Collecting numpy~=1.26.4 (from -r /opt/ai-lab/llama.cpp/requirements/requirements-convert_legacy_llama.txt (line 1))
  Downloading numpy-1.26.4.tar.gz (15.8 MB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 15.8/15.8 MB 17.6 MB/s eta 0:00:00
  Installing build dependencies ... done
  Getting requirements to build wheel ... done
  Installing backend dependencies ... done
```

// encountered issues, but resolved, generally installed missing pythons apps/packages

```
      ../meson.build:44:2: ERROR: Problem encountered: Cannot compile `Python.h`. Perhaps you need to install python-dev|python-d
evel

      A full log can be found at /tmp/pip-install-ep4j7h5w/numpy_6d98eea7bf80429ea18c32684cc78419/.mesonpy-d2rrtgb4/meson-logs/me
son-log.txt
      [end of output]

  note: This error originates from a subprocess, and is likely not a problem with pip.

[notice] A new release of pip is available: 25.0.1 -> 26.0.1
[notice] To update, run: pip install --upgrade pip
error: metadata-generation-failed

× Encountered error while generating package metadata.
╰─> See above for output.
```

```
[notice] A new release of pip is available: 25.0.1 -> 26.0.1
[notice] To update, run: pip install --upgrade pip
sles16-node3:/opt/ai-lab/llama.cpp # pip install --upgrade pip
DEPRECATION: Loading egg at /usr/lib/python3.13/site-packages/suse_lifecycle-0.9-py3.13.egg is deprecated. pip 25.1 will enforce
this behaviour change. A possible replacement is to use pip for package installation. Discussion can be found at https://github.c
om/pypa/pip/issues/12330
Requirement already satisfied: pip in /usr/lib/python3.13/site-packages (25.0.1)
Collecting pip
  Downloading pip-26.0.1-py3-none-any.whl.metadata (4.7 kB)
Downloading pip-26.0.1-py3-none-any.whl (1.8 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1.8/1.8 MB 5.9 MB/s eta 0:00:00
Installing collected packages: pip
  Attempting uninstall: pip
    Found existing installation: pip 25.0.1
```

```
sles16-node3:/opt/ai-lab/llama.cpp # zypper search python-dev
Refreshing service 'openSUSE'.
Loading repository data...
Reading installed packages...

S  | Name                          | Summary                                  | Type
---+-------------------------------+------------------------------------------+--------
   | python313-dbus-python-devel   | Python bindings for D-Bus -- deve->      | package
   | tree-sitter-python-devel      | Devel package for tree-sitter-pyt->      | package

    Note: For an extended search including not yet activated remote resources
    please use 'zypper search-packages'.
sles16-node3:/opt/ai-lab/llama.cpp # zypper install python313-dbus-python-devel
Refreshing service 'openSUSE'.
Loading repository data...
Reading installed packages...
Resolving package dependencies...

The following 17 NEW packages are going to be installed:
  dbus-1-devel dbus-1-glib-bash-completion dbus-1-glib-devel dbus-1-glib-tool
  glib2-devel libffi-devel libgthread-2_0-0 libmount-devel libpcre2-posix3
  libselinux-devel libsepol-devel pcre2-devel python313-dbus-python-devel
  python313-devel python-dbus-python-common-devel typelib-1_0-GIRepository-3_0
  typelib-1_0-GLibUnix-2_0

17 new packages to install.
```

// installed openssl so the cmake rebuild would have https support (that's another certificate rabbit hole)

```
-- ggml commit:  2b10b6267
-- Could NOT find OpenSSL, try to set the path to OpenSSL root folder
RYPTO_LIBRARY OPENSSL_INCLUDE_DIR)
CMake Warning at vendor/cpp-httplib/CMakeLists.txt:150 (message):
  OpenSSL not found, HTTPS support disabled
```

```
sles16-node1:/opt/ai-lab/llama.cpp # zypper install openssl
Refreshing service 'openSUSE'.
Loading repository data...
Reading installed packages...
Resolving package dependencies...

The following 2 NEW packages are going to be installed:
  openssl openssl-3

2 new packages to install.
```

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// actual program to install is the libopenssl-devel

```
sles16-node1:/opt/ai-lab/llama.cpp # zypper install libopenssl-devel
Refreshing service 'openSUSE'.
Loading repository data...
Reading installed packages...
Resolving package dependencies...

The following 3 NEW packages are going to be installed:
  jitterentropy-devel libopenssl-3-devel libopenssl-devel

3 new packages to install.
```

// this was the first command to build the rpc function

```
cmake -B build -DGGML_RPC=ON
```

// this was the second command, both of these were part of the configuration build for llama.cpp for the distribution configuration

```
sles16-node1:/opt/ai-lab/llama.cpp # cmake --build build --config Release -j$(nproc)
[  0%] Building C object examples/gguf-hash/CMakeFiles/sha256.dir/deps/sha256/sha256.c.o
[  0%] Building CXX object vendor/cpp-httplib/CMakeFiles/cpp-httplib.dir/httplib.cpp.o
[  0%] Building C object ggml/src/CMakeFiles/ggml-base.dir/ggml.c.o
[  0%] Building C object ggml/src/CMakeFiles/ggml-base.dir/ggml-alloc.c.o
[  0%] Building C object examples/gguf-hash/CMakeFiles/sha1.dir/deps/sha1/sha1.c.o
[  0%] Building CXX object common/CMakeFiles/build_info.dir/build-info.cpp.o
[  0%] Building CXX object ggml/src/CMakeFiles/ggml-base.dir/ggml.cpp.o
[  0%] Building CXX object tools/mtmd/CMakeFiles/llama-llava-cli.dir/deprecation-warning.cpp.o
[  1%] Building C object examples/gguf-hash/CMakeFiles/xxhash.dir/deps/xxhash/xxhash.c.o
[  1%] Building CXX object ggml/src/CMakeFiles/ggml-base.dir/ggml-opt.cpp.o
[  2%] Building CXX object tools/mtmd/CMakeFiles/llama-minicpmv-cli.dir/deprecation-warning.cpp.o
[  4%] Building CXX object ggml/src/CMakeFiles/ggml-base.dir/ggml-backend.cpp.o
[  4%] Building CXX object tools/mtmd/CMakeFiles/llama-gemma3-cli.dir/deprecation-warning.cpp.o
[  4%] Building CXX object tools/mtmd/CMakeFiles/llama-qwen2vl-cli.dir/deprecation-warning.cpp.o
[  4%] Built target build_info
```

// if the service can't connect, double check the firewall and add the ports

```
sles16-node1:/opt/ai-lab/llama.cpp # firewall-cmd --add-port=50052/tcp --permanent
success
sles16-node1:/opt/ai-lab/llama.cpp # firewall-cmd --reload
success
```

// "firewall-cmd –add-port=8080/tcp –permananent" may also be needed for the open webui

// check the sshd service, if trying to scp some files between the nodes it may be off by default on suse 16

```
sles16-node1:/opt/ai-lab/llama.cpp # systemctl status sshd
○ sshd.service - OpenSSH Daemon
     Loaded: loaded (/usr/lib/systemd/system/sshd.service; disabled; preset: di>
     Active: inactive (dead)
```

```
sles16-node1:/opt/ai-lab/llama.cpp # systemctl enable sshd
Created symlink '/etc/systemd/system/multi-user.target.wants/sshd.service' → '/usr/lib/systemd/system/sshd.service'.
sles16-node1:/opt/ai-lab/llama.cpp # systemctl restart sshd
```

// launching the single server instance, rather than the cli instance

```
sles16-node1:/opt/ai-lab/llama.cpp # build/bin/llama-server -m ../../ai-lab/gemma3-27b-itqat.gguf --host 0.0.0.0 --port 8080 --th
reads 14 --alias "Node 1 solo"
main: n_parallel is set to auto, using n_parallel = 4 and kv_unified = true
build: 8212 (2b10b6267) with GNU 15.2.0 for Linux x86_64
system info: n_threads = 14, n_threads_batch = 14, total_threads = 14

system_info: n_threads = 14 (n_threads_batch = 14) / 14 | CPU : SSE3 = 1 | SSSE3 = 1 | AVX = 1 | AVX2 = 1 | F16C = 1 | FMA = 1 |
BMI2 = 1 | AVX512 = 1 | LLAMAFILE = 1 | OPENMP = 1 | REPACK = 1 |

Running without SSL
init: using 13 threads for HTTP server
start: binding port with default address family
main: loading model
srv    load_model: loading model '../../ai-lab/gemma3-27b-itqat.gguf'
common_init_result: fitting params to device memory, for bugs during this step try to reproduce them with -fit off, or provide --
verbose logs if the bug only occurs with -fit on
llama_params_fit_impl: no devices with dedicated memory found
llama_params_fit: successfully fit params to free device memory
llama_params_fit: fitting params to free memory took 0.85 seconds
llama_model_loader: loaded meta data with 47 key-value pairs and 808 tensors from ../../ai-lab/gemma3-27b-itqat.gguf (version GGU
F V3 (latest))
llama_model_loader: Dumping metadata keys/values. Note: KV overrides do not apply in this output.
llama_model_loader: - kv   0:                       general.architecture str              = gemma3
llama_model_loader: - kv   1:                               general.type str              = model
llama_model_loader: - kv   2:                               general.name str              = Gemma 3 27b It
llama_model_loader: - kv   3:                           general.finetune str              = it
llama_model_loader: - kv   4:                           general.basename str              = gemma-3
```

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// this was trying to convert the llm instance that is usually broken up into one file to run it as the .gguf file, but difficulties that weren't resolved via this method, so the curl method was used to pull a working one from the web

// the convert script was native with the python package

```
sles16-node3:/opt/ai-lab/llama.cpp # python3 convert_hf_to_gguf.py ../../ai-lab/unified-blob-hash/ --outfile ../../ai-lab/unified
-gemma3-27b-itqat.gguf
INFO:hf-to-gguf:Loading model: unified-blob-hash
WARNING:hf-to-gguf:Failed to load model config from ../../ai-lab/unified-blob-hash: Unrecognized model in ../../ai-lab/unified-bl
ob-hash. Should have a `model_type` key in its config.json, or contain one of the following strings in its name: aimv2, aimv2_vis
ion_model, albert, align, altclip, apertus, arcee, aria, aria_text, audio-spectrogram-transformer, autoformer, aya_vision, bamba,
 bark, bart, beit, bert, bert-generation, big_bird, bigbird_pegasus, biogpt, bit, bitnet, blenderbot, blenderbot-small, blip, bli
p-2, blip_2_qformer, bloom, blt, bridgetower, bros, camembert, canine, chameleon, chinese_clip, chinese_clip_vision_model, clap,
```

// the following shows attempts to try to address the issues by renaming the files to the requested/expected files of config.json and tokenizer.json, but things were just to new or too broken to work

```
FileNotFoundError: [Errno 2] No such file or directory: '../../ai-lab/unified-blob-hash/config.json'
sles16-node3:/opt/ai-lab/llama.cpp # grep -l "model_type" ../../ai-lab/unified-blob-hash/
grep: ../../ai-lab/unified-blob-hash/: Is a directory
sles16-node3:/opt/ai-lab/llama.cpp # grep -l "model_type" ../../ai-lab/unified-blob-hash/*
../../ai-lab/unified-blob-hash/sha256-870e9c20a913cbec51bee4a59f0747ee39e0a0ee2cd8cfc26f6ad6ba9681a7fe
sles16-node3:/opt/ai-lab/llama.cpp # grep -l "tokenizer" ../../ai-lab/unified-blob-hash/*

../../ai-lab/unified-blob-hash/sha256-ccc0cddac56136ef0969cf2e3e9ac051124c937be42503b47ec570dead85ff87
sles16-node3:/opt/ai-lab/llama.cpp # cp -v ../../ai-lab/unified-blob-hash/sha256-870e9c20a913cbec51bee4a59f0747ee39e0a0ee2cd8cfc2
6f6ad6ba9681a7fe ../../ai-lab/unified-blob-hash/config.json
'../../ai-lab/unified-blob-hash/sha256-870e9c20a913cbec51bee4a59f0747ee39e0a0ee2cd8cfc26f6ad6ba9681a7fe' -> '../../ai-lab/unified
-blob-hash/config.json'
sles16-node3:/opt/ai-lab/llama.cpp # cp -v ../../ai-lab/unified-blob-hash/sha256-ccc0cddac56136ef0969cf2e3e9ac051124c937be42503b4
7ec570dead85ff87 ../../ai-lab/unified-blob-hash/tokenizer.json
'../../ai-lab/unified-blob-hash/sha256-ccc0cddac56136ef0969cf2e3e9ac051124c937be42503b47ec570dead85ff87' -> '../../ai-lab/unified
-blob-hash/tokenizer.json'
```

// part of the troubleshooting, emphasis on the pip method, it had to be the "python3 -m"

```
sles16-node3:/opt/ai-lab/llama.cpp # pip install --upgrade transformers
bash: /bin/pip: No such file or directory
sles16-node3:/opt/ai-lab/llama.cpp # pip install git+https://github.com/huggingface/transformers.git
bash: /bin/pip: No such file or directory
sles16-node3:/opt/ai-lab/llama.cpp # python3 -m pip install --upgrade transformers
Requirement already satisfied: transformers in /usr/local/lib/python3.13/site-packages (4.57.6)
Collecting transformers
  Downloading transformers-5.3.0-py3-none-any.whl.metadata (32 kB)
Collecting huggingface-hub<2.0,>=1.3.0 (from transformers)
  Downloading huggingface_hub-1.5.0-py3-none-any.whl.metadata (13 kB)
Requirement already satisfied: numpy>=1.17 in /usr/local/lib64/python3.13/site-packages (from transformers) (1.26.4)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.13/site-packages (from transformers) (26.0)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib64/python3.13/site-packages (from transformers) (6.0.3)
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib64/python3.13/site-packages (from transformers) (2026.2.28)
```

// using curl to get the working instance of the .gguf file

```
sles16-node3:/opt/ai-lab/unified-blob-hash # curl -L -o ../../ai-lab/unified-blob-hash/gemma3-27b-itqat.gguf https://huggingface.
co/mradermacher/gemma-3-27b-it-GGUF/resolve/main/gemma-3-27b-it.Q4_K_M.gguf
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100  1361  100  1361    0     0   1484      0 --:--:-- --:--:-- --:--:--  1484
100 15.4G  100 15.4G    0     0  38.4M      0  0:06:49  0:06:49 --:--:-- 39.1M
```

// starting the llama-cli instance (distributed dual brain configuration)

```
sles16-node3:/opt/ai-lab/unified-blob-hash # ../../ai-lab/llama.cpp/build/bin/llama-cli -m ../../ai-lab/unified-blob-hash/gemma3-
27b-itqat.gguf --rpc 172.[redacted]:50052,172[redacted]:50052 --n-gpu-layers 0

Loading model...

llama.cpp

build      : b8212-2b10b6267
model      : gemma3-27b-itqat.gguf
modalities : text

available commands:
  /exit or Ctrl+C     stop or exit
  /regen              regenerate the last response
  /clear              clear the chat history
  /read               add a text file
```

// testing the cli instance prompt

```
> hello

Hello there! 👋

How can I help you today? Just let me know what you're thinking, or if you just wanted to say hi, that's great too!

I can:

* **Answer questions:** About pretty much anything!
* **Generate creative content:** Like stories, poems, code, scripts, musical pieces, email, letters, etc.
* **Translate languages.**
* **Summarize text.**
* **Help with brainstorming.**
* **Just chat!**



[ Prompt: 2.6 t/s | Generation: 1.0 t/s ]
```


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// exiting the instance

```
>

Exiting...
llama_memory_breakdown_print: | memory breakdown [MiB]          |  total     free     self   model   context   compute    unaccoun
ted |
llama_memory_breakdown_print: |   - RPC0 (172.16.11.188:50052) | 257738 = 257738 + (    0 =     0 +       0 +       0) +
  0 |
llama_memory_breakdown_print: |   - RPC1 (172.16.11.189:50052) | 257738 = 257738 + (    0 =     0 +       0 +       0) +
  0 |
llama_memory_breakdown_print: |   - Host                        |                   27170 = 15773 +   10864 +     533
    |
llama_memory_breakdown_print: |   - CPU_REPACK                  |                   11694 = 11694 +       0 +       0
    |
sles16-node3:/opt/ai-lab/unified-blob-hash #
```

// launching the server instance (distributed dual brain configuration)

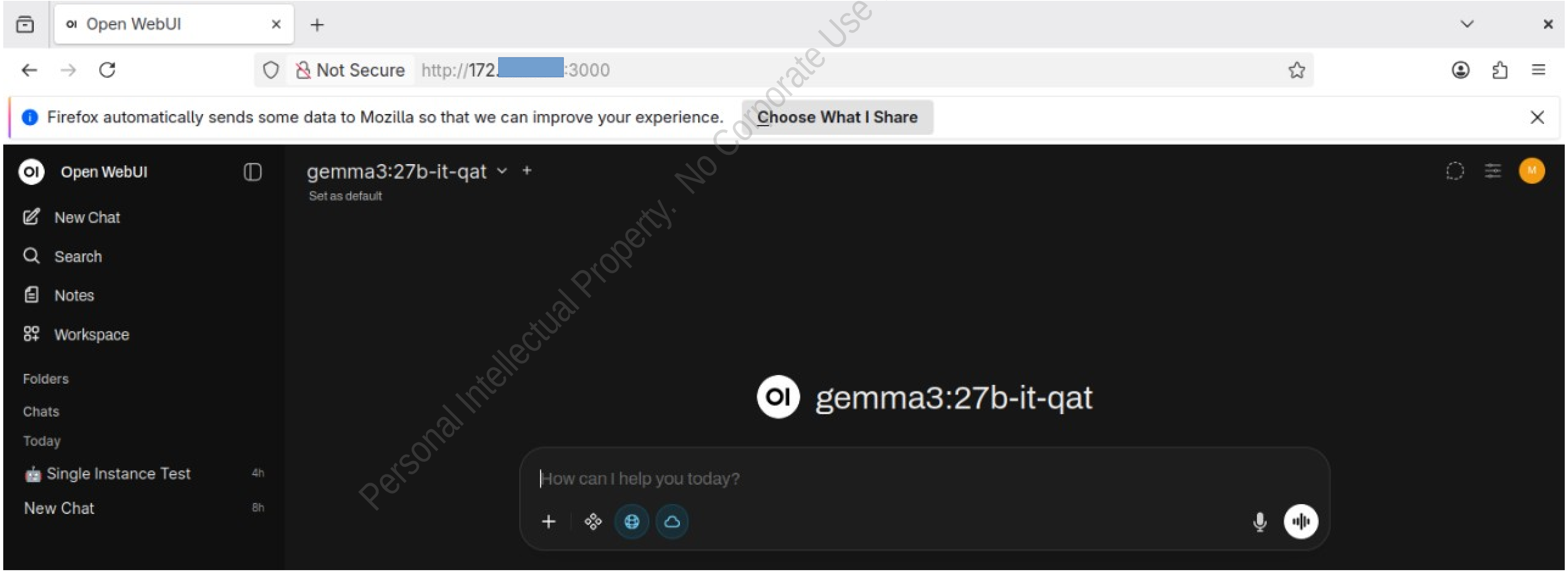

```
sles16-node3:/opt/ai-lab/unified-blob-hash # ../../ai-lab/llama.cpp/build/bin/llama-server -m ../../ai-lab/unified-blob-hash/gemm
a3-27b-itqat.gguf --rpc 172.[redacted]50052,172[redacted]50052 --host 0.0.0.0 --port 8080 --n-gpu-layers 0

main: n_parallel is set to auto, using n_parallel = 4 and kv_unified = true
build: 8212 (2b10b6267) with GNU 15.2.0 for Linux x86_64
system info: n_threads = 8, n_threads_batch = 8, total_threads = 8

system_info: n_threads = 8 (n_threads_batch = 8) / 8 | CPU : SSE3 = 1 | SSSE3 = 1 | AVX = 1 | AVX2 = 1 | F16C = 1 | FMA = 1 | BMI
2 = 1 | AVX512 = 1 | LLAMAFILE = 1 | OPENMP = 1 | REPACK = 1 |

Running without SSL
init: using 7 threads for HTTP server
start: binding port with default address family
main: loading model
srv    load_model: loading model '../../ai-lab/unified-blob-hash/gemma3-27b-itqat.gguf'
common_init_result: fitting params to device memory, for bugs during this step try to reproduce them with -fit off, or provide --
verbose logs if the bug only occurs with -fit on
llama_params_fit_impl: projected memory use with initial parameters [MiB]:
llama_params_fit_impl:   - RPC0 (172.16.11.188:50052): 257738 total,      0 used, 257738 free vs. target of   1024
llama_params_fit_impl:   - RPC1 (172.16.11.189:50052): 257738 total,      0 used, 257738 free vs. target of   1024
llama_params_fit_impl: projected to use 0 MiB of device memory vs. 515477 MiB of free device memory
llama_params_fit_impl: targets for free memory can be met on all devices, no changes needed
llama_params_fit: successfully fit params to free device memory
llama_params_fit: fitting params to free memory took 1.71 seconds
llama_model_load_from_file_impl: using device RPC0 (172.16.11.188:50052) (unknown id) - 257738 MiB free
llama_model_load_from_file_impl: using device RPC1 (172.16.11.189:50052) (unknown id) - 257738 MiB free
llama_model_loader: loaded meta data with 47 key-value pairs and 808 tensors from ../../ai-lab/unified-blob-hash/gemma3-27b-itqat
```

// attempted to install the open-webui interface for the ai llm but it complained about python version 3.11 not existing, so went the docker route

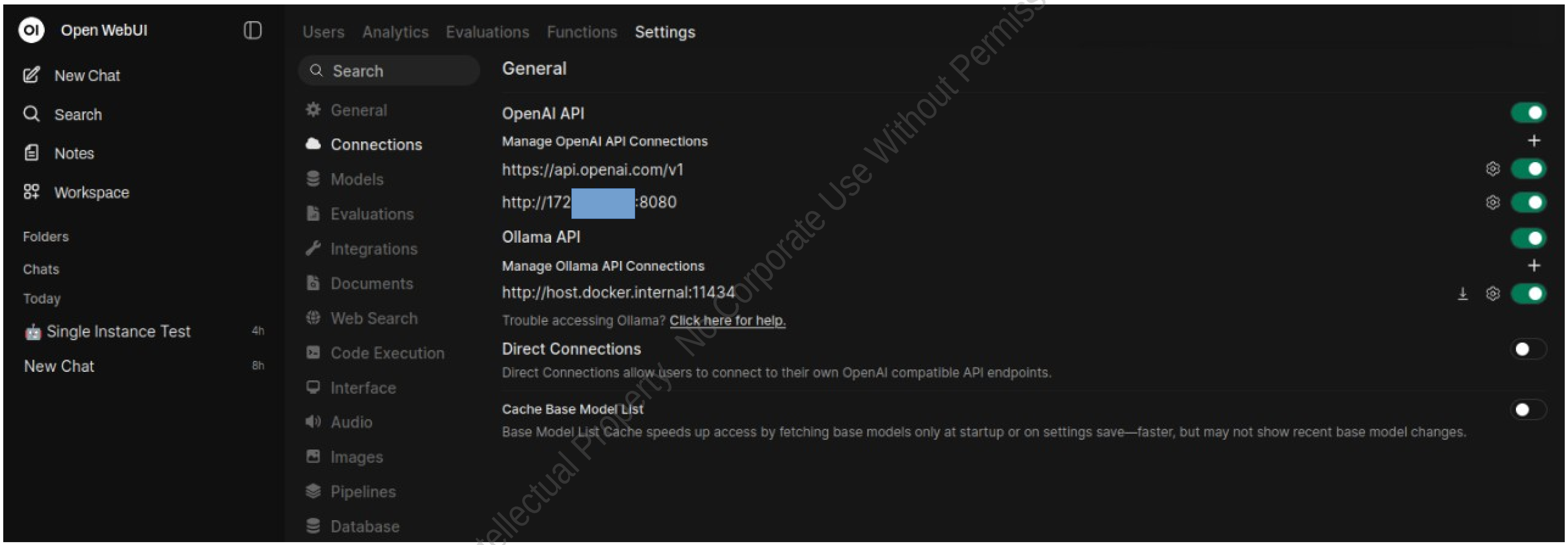
```
ERROR: Could not find a version that satisfies the requirement open-webui (from versions: none)
ERROR: No matching distribution found for open-webui
sles16-node3:/opt/ai-lab/unified-blob-hash # python3 --version
Python 3.13.12
sles16-node3:/opt/ai-lab/unified-blob-hash # zypper install docker
Refreshing service 'openSUSE'.
Loading repository data...
Reading installed packages...
Resolving package dependencies...

The following 2 recommended packages were automatically selected:
  criu docker-rootless-extras

The following 16 NEW packages are going to be installed:
  catatonit containerd criu docker docker-bash-completion docker-buildx
  docker-rootless-extras fuse-overlayfs iptables libnet9 libslirp0
  python313-protobuf rootlesskit runc slirp4netns xtables-plugins

16 new packages to install.
```

// pulled, installed, started docker for open-webui installation

```
sles16-node3:/opt/ai-lab/unified-blob-hash # systemctl start docker
sles16-node3:/opt/ai-lab/unified-blob-hash #
sles16-node3:/opt/ai-lab/unified-blob-hash # docker run -d -p 3000:8080 --add-host=host.docker.internal:host-gateway -v open-webu
i:/app/backend/data --name open-webui ghcr.io/open-webui/open-webui:main
Unable to find image 'ghcr.io/open-webui/open-webui:main' locally
main: Pulling from open-webui/open-webui
84a2afebaf4d: Pull complete
0ca4ca8e967f: Pull complete
4afda7e4a767: Pull complete
bb97a7af1971: Pull complete
```



// the configurations shows a default instance and an external instance the stand alone configuration



// standard server instance of the gemma3 model with the built in web ui, not the open-webui

// this active cluster - 400 series